

Vishwambhara R Hebbalalu

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Profile

Computer Science student at PES University and IIT Madras with experience spanning software engineering, machine learning, data pipelines, and systems development.

Experienced in building end-to-end applications and research prototypes across web, AI/ML, robotics, and embedded environments.

Interested in solving practical problems through a combination of strong programming fundamentals, experimentation, and system-level thinking.



Education

PES University | (2023 - present)

Bachelor of Technology in **Computer Science and Engineering**

IIT Madras | (2023 - present)

Bachelor of Science in **Data Science**

Experience

Teaching Assistant | Software Engineering - PESU (August-December 2026)

Handling lab curriculum and demonstrations for the subject of Software Engineering, taken by junior year (5th semester) students.

Intern | MARS Labs - IoT PESU (June-August 2026)

Worked on a cardiac simulation pipeline, from an ECG to a WebGL simulation of the human heart. Handled machine-learning based ECG delineation and ECG artifact detection.

Research Intern | Saraswatam (June-August 2025)

Engaged in a research-driven internship in computational linguistics with a focus on classical Indic languages. Contributed to the design and implementation of methodologies for text digitization and interlingual transformation. The work emphasized linguistic accuracy, script-level processing, and the preservation of philological structure.

Web Developer | GCube (April-October 2025)

Developed and maintained web applications using React/Vite as part of the club's web development team.

Publications and Preprints

- Physiologically Guided R-Centered Deep Learning for ECG Wave Delineation across Heterogeneous Databases, (accepted at the 13th International Conference on Biomedical and Bioinformatics Engineering, Hokkaido)
- Dual-Stage OCR Correction for Classical Languages using LLMs: A Comparative Evaluation, ResearchGate preprint (has been cited)

Skills

- Programming Languages and Frameworks:
 - Languages: Python, C, JavaScript, Java, Assembly, Verilog
 - ML/Data: Tensorflow, PyTorch
 - Systems/Robotics: ROS2 Humble, Jazzy
 - Web: React, Node.js, TypeScript

- Machine Learning:
 - *Fundamentals of Reinforcement Learning* organized by PESU I/O
 - *Artificial Intelligence and Machine Learning* organized by Pravega, IISc
 - *Foundational Course in Data Science* by IIT Madras
- Part of Web Development team at GCube, game development club of PESU

Projects

Visualizing ECG-Driven Cardiac Motion (Internship 2026)

- Cardiac simulation pipeline using ECG-derived events as common timing references for multiple representations of the heart.
- Conference paper accepted on a sub-part of the project: R-peak centered ECG Delineation

ROS2-Based Maze Following Robot (Simulation)

- Developed a maze-following robot using ROS2 with modular node-based architecture
- Implemented sensor-driven control logic for navigation using real-time data streams
- Utilized publisher-subscriber communication for inter-node messaging and system coordination
- Focused on reliable behavior in constrained and dynamic environments

Data Pipeline for Agricultural Decision Support System (Terrathon 4.0 Winner)

- Built data ingestion and preprocessing pipelines using web scraping from multiple agricultural sources
- Cleaned and structured heterogeneous datasets for downstream model training
- Contributed to frontend integration and system-level workflow design

Web Application for Classical Literature (Kalidasa)

- Developed a web platform for accessing and presenting structured literary content
- Focused on clean data representation and frontend performance

Lightweight Container Runtime

- Implementing a minimal container runtime with layered filesystem extraction
- Exploring namespace-based isolation and process-level resource control
- Focused on understanding OS-level abstractions and execution environments

Awards and Achievements

- Team awarded 1st Prize in Terrathon 4.0, a 24hr hackathon organized by The Alcoding Club and The Changemakers' Society, sponsored by Cisco and KarLex AI.
- First Prize in Space Quiz organized by Equinox PESU.
- Awarded best short story by the Chinmaya Mission in the Wordscapes competition, about the Ramayana.